

CDMA - BASIC

Imagine a scenario where a sender and receiver using CDMA (Section 7.2.1 from the text) are exchanging data. Assume both the sender and receiver use a CDMA code of size 6, where $M = 1, 1, 1, -1, -1, 1$.

SOLUTION

1. The encoded output is: 1,1,1,-1,-1,1
2. The original bit value is: 1

QUESTION 1 OF 2

Given the CDMA code above and the bit $d=1$, what is the encoded output? Separate each value with a comma and no spaces

Answer

CDMA - BASIC

Imagine a scenario where a sender and receiver using CDMA (Section 7.2.1 from the text) are exchanging data. Assume both the sender and receiver use a CDMA code of size 9, where $M = -1, 1, 1, 1, 1, -1, 1, 1, 1$.

SOLUTION

1. The encoded output is: 1,-1,-1,-1,-1,1,-1,-1,-1
2. The original bit value is: -1

QUESTION 1 OF 2

Given the CDMA code above and the bit $d=-1$, what is the encoded output? Separate each value with a comma and no spaces

Answer

CDMA - BASIC

Imagine a scenario where a sender and receiver using CDMA (Section 7.2.1 from the text) are exchanging data. Assume both the sender and receiver use a CDMA code of size 7, where $M = 1, 1, 1, -1, -1, 1, -1$.

SOLUTION

1. The encoded output is: -1,-1,-1,1,1,-1,1
2. The original bit value is: -1

QUESTION 1 OF 2

Given the CDMA code above and the bit $d=-1$, what is the encoded output? Separate each value with a comma and no spaces

Answer

CDMA - BASIC

Imagine a scenario where a sender and receiver using CDMA (Section 7.2.1 from the text) are exchanging data. Assume both the sender and receiver use a CDMA code of size 8, where $M = 1, 1, -1, 1, -1, 1, 1, -1$.

SOLUTION

1. The encoded output is: -1,-1,1,-1,1,-1,-1,1
2. The original bit value is: 1

QUESTION 1 OF 2

Given the CDMA code above and the bit $d=-1$, what is the encoded output? Separate each value with a comma and no spaces

Answer

CDMA - ADVANCED

This time, assume there are 2 senders whom interfere with each other and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of $(-1, -1, -1, 1, 1, 1)$ and sender 2 has a CDMA code of $(1, -1, -1, -1, -1, 1)$ and their combined output is: $(-2, 0, 0, 2, 2, 0)$

SOLUTION

1. The decoded bit is: 1 for sender 1
2. The decoded bit is: -1 for sender 2

QUESTION 1 OF 2

Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

Answer

CDMA - ADVANCED

This time, assume there are 2 senders whom interfere with each other and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of (1, -1, -1, 1, 1, 1, -1, -1) and sender 2 has a CDMA code of (-1, -1, 1, 1, -1, 1, 1, -1) and their combined output is: (0, 2, 0, -2, 0, -2, 0, 2)

SOLUTION

1. The decoded bit is: -1 for sender 1
2. The decoded bit is: -1 for sender 2

QUESTION 1 OF 2

Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

Answer

CDMA - ADVANCED

This time, assume there are 2 senders whom interfere with each other and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of (1, -1, -1, -1, 1, -1) and sender 2 has a CDMA code of (1, 1, 1, -1, -1, 1) and their combined output is: (-2, 0, 0, 2, 0, 0)

SOLUTION

1. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit
2. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit

QUESTION 1 OF 2

Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

Answer

CDMA - ADVANCED

This time, assume there are 3 senders whom interfere with each other and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of (1, -1, 1, -1, -1, -1), sender 2 has a CDMA code of (-1, -1, -1, 1, -1, -1), and sender 3 has a CDMA code of (1, 1, -1, -1, -1, -1) and their combined output is: (-3, -1, -1, 3, 1, 1)

SOLUTION

1. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit
2. The decoded bit is: 1 for sender 2
3. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit

QUESTION 1 OF 3

Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

Answer

CDMA - ADVANCED

This time, assume there are 3 senders whom interfere with each other and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of $(-1, 1, 1, 1, -1, 1, -1, -1)$, sender 2 has a CDMA code of $(-1, 1, -1, -1, 1, -1, -1, -1)$, and sender 3 has a CDMA code of $(-1, -1, 1, -1, 1, -1, -1, 1)$ and their combined output is: $(-1, 3, -1, 1, -1, 1, -1, -3)$

SOLUTION

1. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit
2. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit
3. The decoded bit is: -1 for sender 3

QUESTION 1 OF 3

Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

Answer

CDMA - ADVANCED

This time, assume there are 3 senders whom interfere with each other and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of $(-1, 1, -1, 1, 1, -1)$, sender 2 has a CDMA code of $(1, -1, -1, 1, 1, -1)$, and sender 3 has a CDMA code of $(-1, -1, -1, -1, -1, 1)$ and their combined output is: $(-1, 3, 1, 1, 1, -1)$

SOLUTION

1. The decoded bit is: 1 for sender 1
2. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit
3. The decoded bit is: -1 for sender 3

QUESTION 1 OF 3

Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

Answer